

Ex. Test, which of the following are prime numbers?

(i) 137

(ii) 173

(iii) 319

(iv) 437

(v) 811

- Sol.** (i) We know that $(12)^2 > 137$.
Prime numbers less than 12 are 2, 3, 5, 7, 11.
Clearly, none of them divides 137.
 $\therefore$ 137 is a prime number.
- (ii) We know that $(14)^2 > 173$.
Prime numbers less than 14 are 2, 3, 5, 7, 11, 13.
Clearly, none of them divides 173.
 $\therefore$ 173 is a prime number.
- (iii) We know that $(18)^2 > 319$.
Prime numbers less than 18 are 2, 3, 5, 7, 11, 13, 17.
Out of these prime numbers, 11 divides 319 completely.
 $\therefore$ 319 is not a prime number.
- (iv) We know that $(21)^2 > 437$.
Prime numbers less than 21 are 2, 3, 5, 7, 11, 13, 17, 19.
Clearly, 437 is divisible by 19.
 $\therefore$ 437 is not a prime number.
- (v) We know that $(30)^2 > 811$.
Prime numbers less than 30 are 2, 3, 5, 7, 11, 13, 17, 19, 23, 29.
Clearly, none of these numbers divides 811.
 $\therefore$ 811 is a prime number.

V. Important Formulae:

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| (i) $(a + b)^2 = a^2 + b^2 + 2ab$ | (ii) $(a - b)^2 = a^2 + b^2 - 2ab$ |
| (iii) $(a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$ | (iv) $(a + b)^2 - (a - b)^2 = 4ab$ |
| (v) $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$ | (vi) $(a - b)^3 = a^3 - b^3 - 3ab(a - b)$ |
| (vii) $a^2 - b^2 = (a + b)(a - b)$ | (viii) $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$ |
| (ix) $a^3 + b^3 = (a + b)(a^2 + b^2 - ab)$ | (x) $a^3 - b^3 = (a - b)(a^2 + b^2 + ab)$ |
| (xi) $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$ | |
| (xii) If $a + b + c = 0$, then $a^3 + b^3 + c^3 = 3abc$ | |

TESTS OF DIVISIBILITY

1. Divisibility By 2:

A number is divisible by 2 if its unit digit is any of 0, 2, 4, 6, 8.

Ex. 58694 is divisible by 2, while 86945 is not divisible by 2.

2. Divisibility By 3:

A number is divisible by 3 only when the sum of its digits is divisible by 3.

Ex. (i) In the number 695421, the sum of digits = 27, which is divisible by 3.

$\therefore$ 695421 is divisible by 3.

(ii) In the number 948653, the sum of digits = 35, which is not divisible by 3.

$\therefore$ 948653 is not divisible by 3.

3. Divisibility By 9:

A number is divisible by 9 only when the sum of its digits is divisible by 9.

Ex. (i) In the number 246591, the sum of digits = 27, which is divisible by 9.

$\therefore$ 246591 is divisible by 9.

(ii) In the number 734519, the sum of digits = 29, which is not divisible by 9.

$\therefore$ 734519 is not divisible by 9.

4. Divisibility By 4:

A number is divisible by 4 if the number formed by its last two digits is divisible by 4.

Ex. (i) 6879376 is divisible by 4, since 76 is divisible by 4.

(ii) 496138 is not divisible by 4, since 38 is not divisible by 4.

5. Divisibility By 8:

A number is divisible by 8 if the number formed by its last three digits is divisible by 8.